

This fix didn't work Get INIT Running

1. **Secrets Manager Console** → find `lab/rds/mysql`
2. Click **Retrieve secret value**
3. Check if you see a `host` field with your RDS endpoint

If `host` is missing:

- Delete the old secret (or create a new one)
- Create a new secret using "**Credentials for RDS database**"
- Select your `lab-mysql` RDS instance — this auto-fills the host/port/engine
- Name it `lab/rds/mysql`

That should get your `/init` working! 🎯

Based on your lab setup, here's a quick troubleshooting checklist for when the page won't load at all:

3. Is the Flask app actually running?

- SSH into the instance and run:

```
curl http://localhost/list
```

- If that works locally, the app is running — problem is network/SG
- If it errors, the app didn't start — check logs or re-run the bootstrap script

4. Flask binding issue

- The app must bind to `0.0.0.0`, not `127.0.0.1`
- In the user data script, look for something like `app.run(host='0.0.0.0', port=80)`

5. Route table

- Confirm the EC2's subnet has a route `0.0.0.0/0 → igw-xxxxx` (Internet Gateway)

```
#!/bin/bash
dnf update -y
dnf install -y python3-pip
pip3 install flask pymysql boto3

mkdir -p /opt/rdsapp
cat >/opt/rdsapp/app.py <<'PY'
import json
import os
import boto3
import pymysql
from flask import Flask, request

REGION = os.environ.get("AWS_REGION", "us-east-2")
SECRET_ID = os.environ.get("SECRET_ID", "lab/rds/mysql")

secrets = boto3.client("secretsmanager", region_name=REGION)

def get_db_creds():
    resp = secrets.get_secret_value(SecretId=SECRET_ID)
    s = json.loads(resp["SecretString"])
    # When you use "Credentials for RDS database", AWS usually stores:
    # username, password, host, port, dbname (sometimes)
    return s

def get_conn():
    c = get_db_creds()
    host = c["host"]
    user = c["username"]
    password = c["password"]
```

```

    port = int(c.get("port", 3306))
    db = c.get("dbname", "labdb") # we'll create this if it
    doesn't exist
    return pymysql.connect(host=host, user=user, password=password,
    port=port, database=db, autocommit=True)

app = Flask(__name__)

@app.route("/")
def home():
    return """
    <h2>EC2 → RDS Notes App</h2>
    <p>POST /add?note=hello</p>
    <p>GET /list</p>
    """

@app.route("/init")
def init_db():
    c = get_db_creds()
    host = c["host"]
    user = c["username"]
    password = c["password"]
    port = int(c.get("port", 3306))

    # connect without specifying a DB first
    conn = pymysql.connect(host=host, user=user, password=password,
    port=port, autocommit=True)
    cur = conn.cursor()
    cur.execute("CREATE DATABASE IF NOT EXISTS labdb;")
    cur.execute("USE labdb;")
    cur.execute("""
        CREATE TABLE IF NOT EXISTS notes (
            id INT AUTO_INCREMENT PRIMARY KEY,
            note VARCHAR(255) NOT NULL
        );
    """)

```

```

        cur.close()
        conn.close()
        return "Initialized labdb + notes table."

@app.route("/add", methods=["POST", "GET"])
def add_note():
    note = request.args.get("note", "").strip()
    if not note:
        return "Missing note param. Try: /add?note=hello", 400

    conn = get_conn()
    cur = conn.cursor()
    cur.execute("INSERT INTO notes(note) VALUES(%s);", (note,))
    cur.close()
    conn.close()
    return f"Inserted note: {note}"

@app.route("/list")
def list_notes():
    conn = get_conn()
    cur = conn.cursor()
    cur.execute("SELECT id, note FROM notes ORDER BY id DESC;")
    rows = cur.fetchall()
    cur.close()
    conn.close()
    out = "<h3>Notes</h3><ul>"
    for r in rows:
        out += f"<li>{r[0]}: {r[1]}</li>"
    out += "</ul>"
    return out

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=80)
PY

```

```
cat >/etc/systemd/system/rdsapp.service <<'SERVICE'
[Unit]
Description=EC2 to RDS Notes App
After=network.target

[Service]
WorkingDirectory=/opt/rdsapp
Environment=SECRET_ID=lab/rds/mysql
ExecStart=/usr/bin/python3 /opt/rdsapp/app.py
Restart=always

[Install]
WantedBy=multi-user.target
SERVICE

systemctl daemon-reload
systemctl enable rdsapp
systemctl start rdsapp
```